MH1101 Tutorial 7 (Week 8)

Reference: Sections 4.1, 4.2 (Lecture Notes)

1. Using the formal definition of limit, prove that the following sequence converges.

(i) $\left\{\dfrac{1}{3^n}\right\}_{n=1}^{\infty}$

(ii) $\left\{\dfrac{2n^2+3}{n^2+1}\right\}_{n=1}^{\infty}$

(iii) $\lim\limits_{n\to\infty} \dfrac{2n^2-3n}{3n^2+1} = \dfrac{2}{3}$. [AY2021 Sem 2 EXAM]

2. Determine whether the sequence converges or diverges. If it converges, find the limit. Justify your answer.

(i) $a_n = \dfrac{3+5n^2}{n+n^2}$

(ii) $a_n = \dfrac{n^4}{n^3-2n}$

(iii) $a_n = \sqrt{\dfrac{n+1}{9n+1}}$

(iv) $a_n = \dfrac{\cos^2 n}{2^n}$

(v) $a_n = \sqrt[n]{2^{1+3n}}$

(vi) $a_n = \dfrac{4^n}{1+9^n}$

(vii) $a_n = \dfrac{(-1)^n n^3}{n^3+2n^2+1}$

(viii) $a_n = \tan\left(\dfrac{2n\pi}{1+8n}\right)$

(ix) $a_n = \dfrac{\tan^{-1} n}{n}$

(x) $a_n = \sqrt[n]{n}$

3. Given a sequence $\{a_n\}_{n=1}^{\infty}$, prove that (using the formal definition of limit) if the odd subsequence $a_1, a_3, \ldots$ and the even subsequence $a_2, a_4, \ldots$ both converge to the same limit L, then $\{a_n\}$ converges to the limit L.

4. Determine whether the statement is true or false. If it is true, explain why it is true. If it is false, explain why it is false, or give an example to show that it is false.

(a) If $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are divergent, then $\{a_n + b_n\}_{n=1}^{\infty}$ is divergent.

(b) If $\{a_n\}$ is divergent, then $\{|a_n|\}$ is divergent.

(c) If $\{a_n\}$ converges to L (real number) and $\{b_n\}$ converges to 0, the $\{a_n b_n\}$ converges to 0.

5. Use Squeeze Theorem to prove that

$$\lim_{n\to\infty} \sqrt[n]{a} = 1, \quad a > 0.$$

Answer Keys. **2.** (i) 5 (ii) ∞ (iii) $\frac{1}{3}$ (iv) 0 (v) 8 (vi) 0 (vii) diverges (viii) 1 (ix) 0 (x) 1